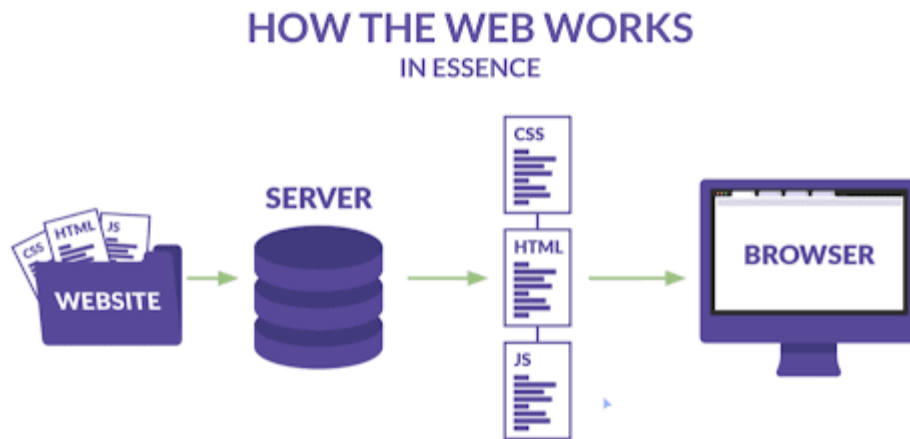
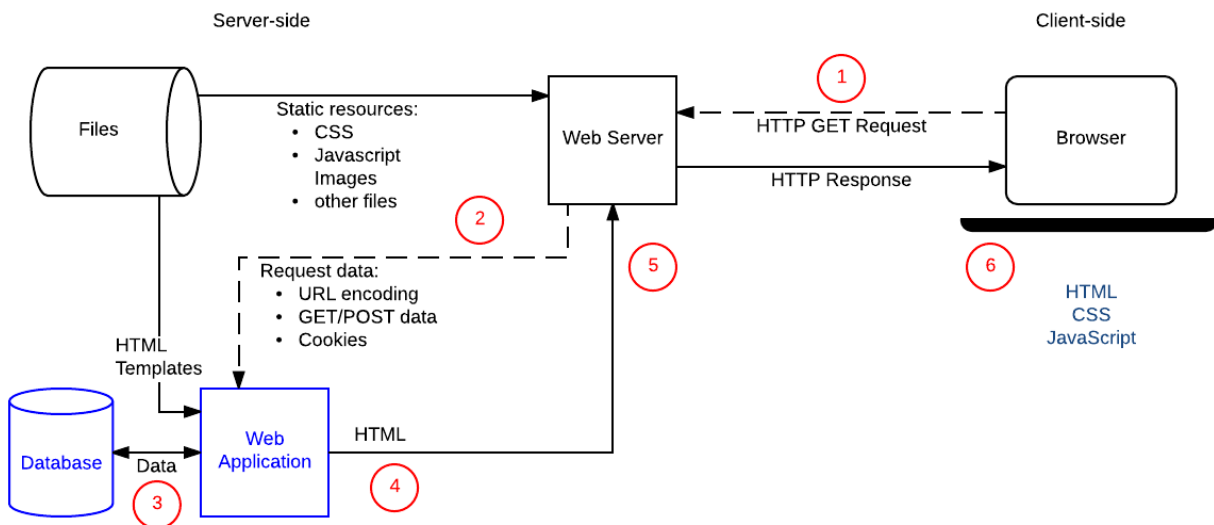


Lesson 3

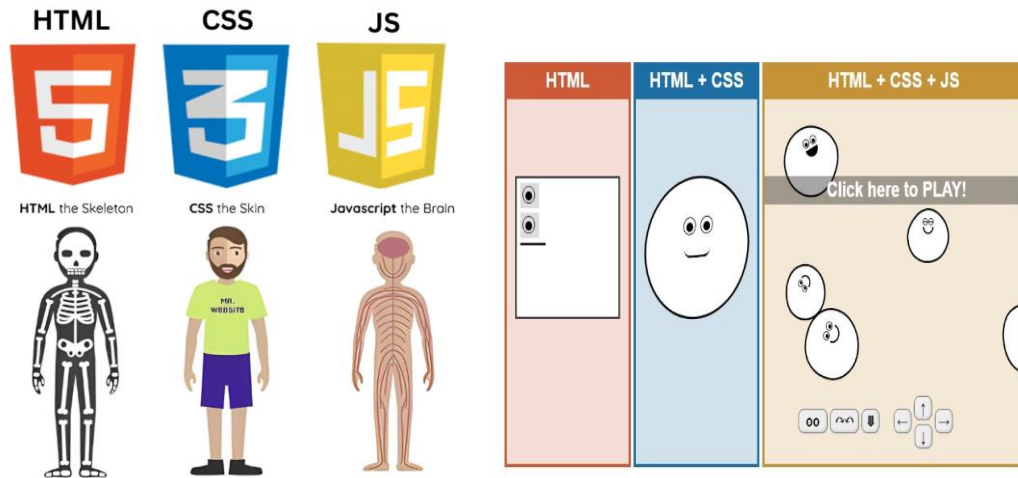
How do websites work? (simple)



How do websites work? (more detailed)

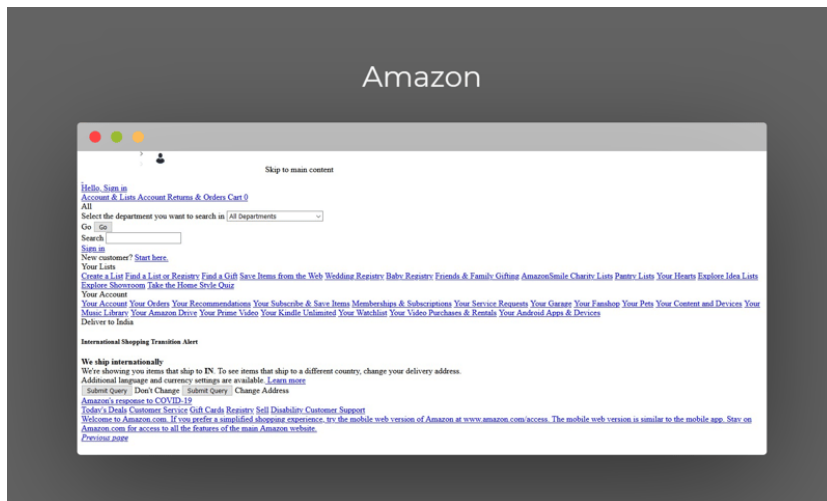


What is HTML, CSS and Javascript?



- **HTML:** the markup language that lays out a page's raw content and structure (headings, paragraphs, images, forms).
- **CSS:** the style sheet language that tells the browser how that content should look (colors, fonts, layout, animations).
- **JavaScript:** the scripting language that adds behavior—responding to clicks, fetching data, updating the page without reloads.
- **Together:** HTML supplies the “what,” CSS the “how it looks,” and JavaScript the “what it can do,” all running in the browser and jointly turning static documents into interactive web apps.

The web without CSS ([amazon.com](https://www.amazon.com) example)



How to host websites



Introduction to streamlit

Fortunately in this course you only need to worry about learning Python. (maybe learning the others can be for a future course)

But now we are getting to the part where you can actually start creating something others can use

I like streamlit as there are tonnes of functions and a developer community so you always can find answers. Web hosting can also be pretty complicated and for your first version of whatever you are building you should just try to get something out there and not focus too much on servers and adding complexity with hosting

Creating a web front-end with just python

1. Download the zip file here and store it in your file directory
2. Open this folder with VS code

3. Let's look at the python file, this time we wrapped inputs in functions for streamlit which allow us to render a user interface.
4. Click **terminal** and **new terminal**
5. Type this code in the terminal **python3 -m venv venv** and hit enter
6. In the terminal at the bottom of the screen type **source venv/bin/activate** and hit **enter**
7. In the terminal type **pip install streamlit** and hit **enter**
8. In the terminal type **streamlit run [beam.py](#)** and hit **enter**
9. Should open a server on <http://localhost:8501/> if it doesn't open automatically then go to this address in your browser
10. Go to Github [here](#) and create an account
11. Click new github repository by clicking **new repository** and list as **public**
12. Back in your VS code, type **git init** in the terminal and hit enter to initialise git in your repo
13. If you get an error message here then make sure you have git installed on your computer, you will otherwise need to download it from [here](#)
14. Type **git add -A** and hit enter (make sure you don't miss the fullstop after add)
15. Type **git commit -m "initial push"** in the terminal and hit enter
16. Copy the URL link from quick setup
17. Type **git remote add origin <the url from your github>**
18. Push the code to github with **git push -u origin main**
19. Create an account with streamlit [here](#)
20. Go back to streamlit and click **create app** and select your repository, this will deploy your app on the web!

Control access

- So when you have built your app chances are you will want to share it with colleagues or clients.
- If your app holds company IP you'd most likely want to make your app private so that only authenticated users can access it.

- You can control access to your app with the following steps:
 - Visit Streamlit
 - Click the dropdown next to your app and click settings
 - Within settings click sharing and edit 'who can view this app' to the setting 'only specific people can view this app'